

EVAC1 part 1

1 Justification of Representation and Algorithm 331 Words

I chose to represent the problem as a single-objective constrained optimisation, where the nutrient and portion constraints are handled with dynamic penalties. Penalties are calculated relative to how hard the restriction is broken i.e. being 5 calories under carries a larger penalty than 1 under. This creates a gradient in the fitness which allows the algorithm to work towards a region with passing solutions, rather than simply failing for breaking constraints. Each individual is represented by a vector of real values, which themselves represent the quantity of each food item in their diet. I chose to represent the food portions as real values, as it closely aligned with the provided constraints; real-valued representation allows the algorithm to explore the search space with the required precision to find an acceptable solution that does not violate any constraints. Also the specification states the solution must have at least 0.5 units of fish, which leads to the assumption that food units cannot be considered as integers.

The $\mu + \lambda$ algorithm was chosen because it is elitist. In a search space with hard constraints, preserving feasible solutions across generations is essential to prevent the loss of high fitness solutions. Also by retaining feasible solutions throughout generations the algorithm can concentrate its search in a smaller search space primarily filled with feasible solutions. A hall of fame is also utilised to ensure we take the very best individual found once the algorithm is complete.

Tournament selection was chosen as it can appropriately handle the potential disparity in fitness created by the penalties. Unlike a fitness proportional solution which can cause a population to stagnate when one individual is significantly better than the rest, the relative ranking of tournament selection maintains a constant selection pressure. Constant selection pressure helps to avoid the case where a single okay individual dominates, as all other solutions fail on constraints and thus have exceptionally high fitness. Blend crossover and Gaussian Mutation were chosen since food portions are continuous.

2 Investigation of Parameters X Words

In order to account for the stochastic nature of genetic algorithms, for each parameter setting 30 independent tests are conducted.

3 Evaluation of Solution X Words

Very good